Welcome To

SECP2523
DATABASE

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INTRODUCTION TO DATABASE

File Based System Approach

- An early attempt to computerize the manual filing system
- Definition:
  - Collection of application programs that perform services for the end users (e.g. reports).
  - Each application program defines and manages its own data.

File Based System Approach



Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

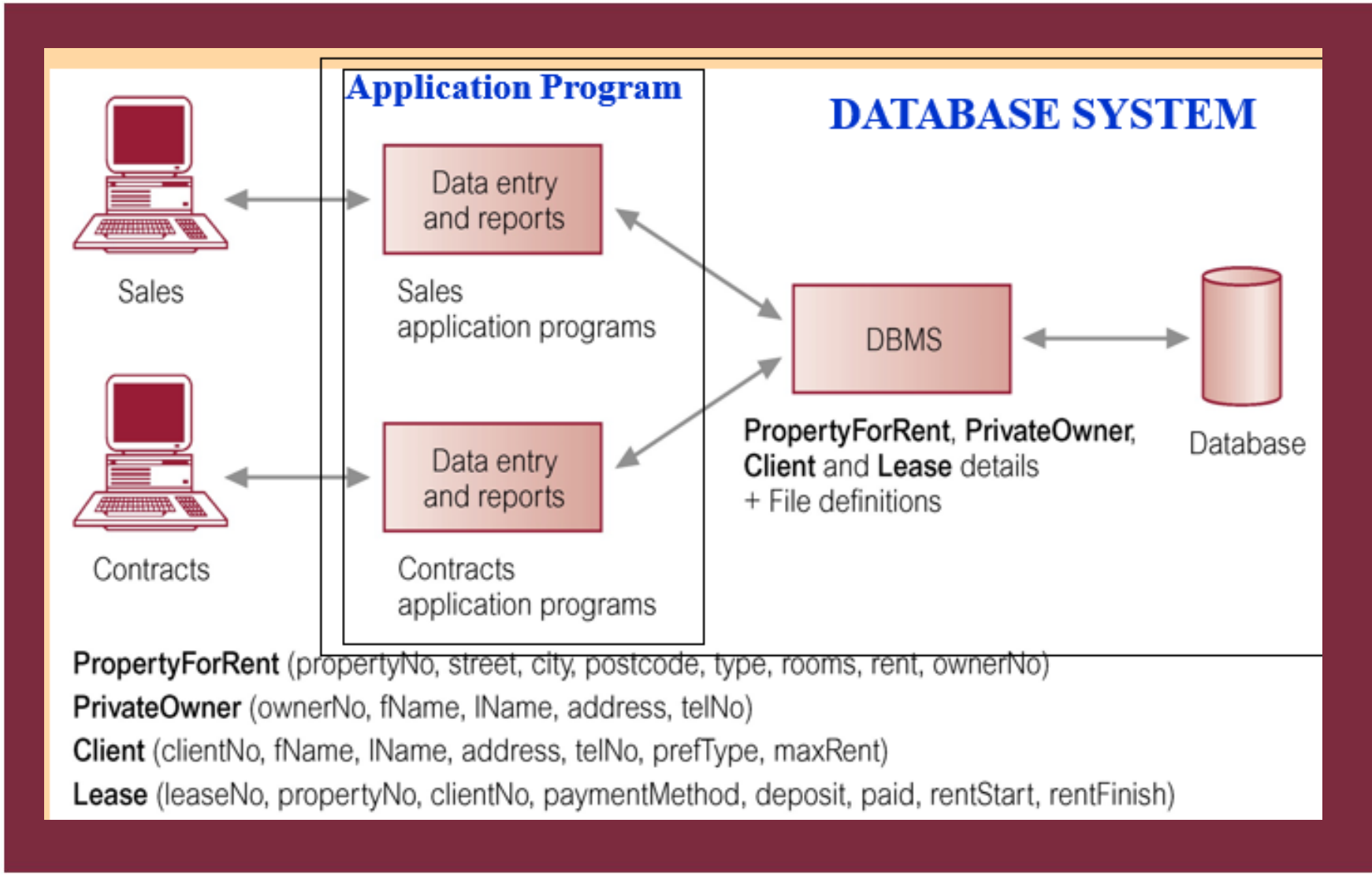
Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

Database System Approach

- database : a collection of related data
- DBMS (database management system) : the software that manages & controls access to the database
- database application : a program that interacts with the database at some point in its execution
- database system : a collection of application programs that interact with the database along with the DBMS and database itself

Database System



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Activity 1

- In a group of 4, identify and discuss 4 possible limitations of traditional file-based approach.
- How do you think the limitations are overcome by a database system? Explain.

RELATIONAL DATABASE

Introduction

- DBMS generations :
  - First → hierarchical, network DBMS – 1960s
  - Second → relational DBMS (RDBMS) – late 70s
  - Third → OODBMS, ORDBMS
  - 4th / 5th → non-relational database (NOSQL)
- RDBMS is based on relational data model concepts by E.F Codd (1970)
  - Simple logical structure
  - Sound theoretical foundation
  - Data is logically structured within relations

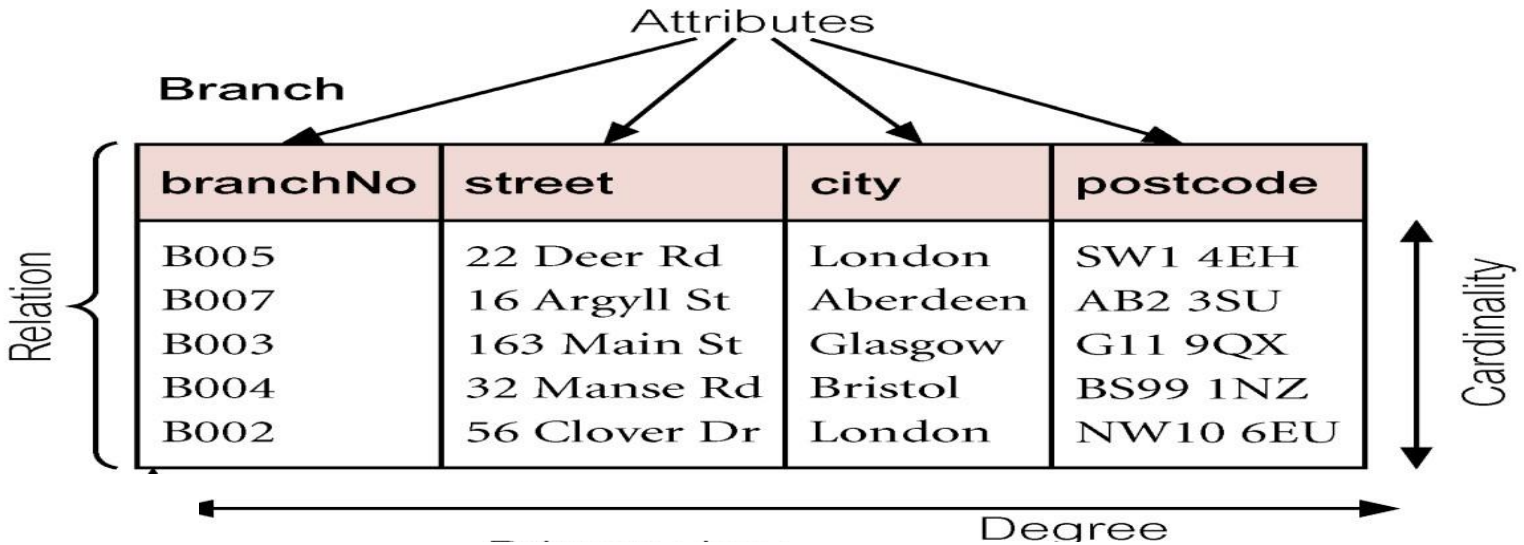
Relational Model Terminology

- A relation is a table with columns and rows.
  - Only applies to logical structure of the database, not the physical structure.
- Attribute is a named column of a relation.
- Domain is the set of allowable values for one or more attributes.

Examples of Attribute & Domains

Attribute	Domain Name	Meaning	Domain Definition
branchNo	BranchNumbers	The set of all possible branch numbers	character: size 4, range B001–B999
street	StreetNames	The set of all street names in Britain	character: size 25
city	CityNames	The set of all city names in Britain	character: size 15
postcode	Postcodes	The set of all postcodes in Britain	character: size 8
sex	Sex	The sex of a person	character: size 1, value M or F
DOB	DatesOfBirth	Possible values of staff birth dates	date, range from 1-Jan-20, format dd-mmm-yy
salary	Salaries	Possible values of staff salaries	monetary: 7 digits, range 6000.00–40000.00

Relational Model Terminology



Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Relational Model Terminologies

- Relation schema
  - Named relation defined by a set of attribute and domain name pairs.
  - Common convention:
    - *RelationName (attribute_1, attribute_2,....., attribute_n)*
    - Example:
      - Branch (branchNo, street, city, postcode)
- Relation database schema
  - Set of relation schemas, each with a distinct name.
- Relation instance
  - A tuple at a specific moment of time
  - Eg: (B005, 55 Jln Dobi, Johor Bahru, 80100)

Relational Keys

- One or more attributes in a relation that will uniquely identifies a tuple in a relation
  - Candidate key
  - Composite key
  - Primary Key
  - Alternate key
  - Foreign key

ACTIVITY 2

Attributes

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Relation

Cardinality

Degree

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Relation

Identify the candidate key(s), primary keys, foreign key (if any) and composite keys (if any) for both relations.

VIEW

- Base Relation
  - Named relation corresponding to an entity in conceptual schema, whose tuples are physically stored in database.
- View
  - Dynamic result of one or more relational operations operating on base relations to produce another relation.
  - A virtual relation that does not necessarily actually exist in the database but is produced upon request, at time of request.

INTEGRITY CONSTRAINT

INTEGRITY CONSTRAINTS

- Integrity constraints – to ensure correctness/ accuracy of data
  - Domain constraints; Integrity rules
- Integrity rules
  - Constraints or restrictions that apply to all instances of a database
  - Two rules:
    - Entity integrity
    - Referential integrity
- Null
  - Represents value for an attribute that is currently unknown or not applicable for tuple

INTEGRITY CONSTRAINTS

- Entity Integrity
  - In a base relation, no attribute of a primary key can be null.
- Referential Integrity
  - If foreign key exists in a relation, the foreign key value:
    - must match a candidate key value of some tuple in its home relation; or
    - foreign key value must be wholly null.
- Enterprise Constraints
  - Additional rules specified by users or database administrators.

ACTIVITY 3: The Info

- The following relational schema form part of a database held in a relational database:-

Hotel	(hotelNo, hotelName, city)
Room	(roomNo, hotelNo, type, price)
Booking	(hotelNo, guestNo, dateFrom, dateTo, roomNo)
Guest	(guestNo, guestName, guestAddress)

- where
  - Hotel contains hotel details and hotelNo is the primary key;
  - Room contains room details for each hotel and (roomNo, hotelNo) forms the primary key;
  - Booking contains details of the bookings and (hotelNo, guestNo, dateFrom) forms the primary key;
  - Guest contains guest details and guestNo is the primary key.

ACTIVITY 3: The Questions

1) Identify the foreign keys in this schema. Explain how the entity and referential integrity rules apply to these relations.

2) Produce some sample tables for these relations that observe the relational integrity rules. Suggest some general constraints that would be appropriate for this schema.

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Thank You

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